SC7 Bayes Methods MT26

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Lecture 6: Markov Chain Monte Carlo Methods I

Markov chain Monte Carlo Methods

MCMC is a family of algorithms for simulating $X_0, X_1, X_2, ...$ so that $X_t \sim p$ (or at least X_t converges to p in distribution) for a user-defined target probability distribution p.

MCMC methods are one of the most versatile classes of Monte Carlo algorithms we have, and are in routine use across statistics.

I will set out the principles for the case that Ω, the space of states of X, is finite (and therefore discrete) and assume they hold for the continuous case.

When we work on a computer we round to computer precision. I call this the computer measure. So in practise we always work with a finite set of states.

Markov chains

Let $\{X_t\}_{t=0}^{\infty}$ be a homogeneous Markov chain of random variables on Ω with $X_0 \sim p^{(0)}$ and transition probability

$$P_{i,j} = \mathbb{P}(X_{t+1} = j | X_t = i).$$

Denote by $P_{i,j}^{(n)}$ the n-step transition probabilities

$$P_{i,j}^{(n)} = \mathbb{P}(X_{t+n} = j | X_t = i)$$

and by $p_i^{(t)} = \mathbb{P}(X_t = i)$, with $p^{(t)}$ a row vector.

The transition matrix P is irreducible if and only if, for each pair of states $i, j \in \Omega$ there is $n < \infty$ such that $P_{i,j}^{(n)} > 0$. The Markov chain is aperiodic if $P_{i,i}^{(n)}$ is non zero for sufficiently large n and all $i \in \Omega$.*

*See Norris, J. *Markov Chains* (2012)

The Stationary Distribution and Detailed Balance

Consider a generic "target" distribution $p_i, i \in \Omega$. This is the distribution we will try to sample. For Bayesian inference, the target distribution will be $\theta \sim \pi(\theta|y)$.

The probability mass function (PMF) $p_i, i \in \Omega$, $\sum_{i\in\Omega} p_i = 1$ is a stationary distribution of P if $pP = p$. If $p^{(0)} = p$ then

$$p_j^{(1)} = \sum_{i\in\Omega} p_i^{(0)} P_{i,j} = [pP]_j,$$

so $p_j^{(1)} = p_j$ also. Iterating, $p^{(t)} = p$ for each $t = 1, 2, ...$ in the chain, so the distribution of $X_t \sim p^{(t)}$ doesn't change with t, it is stationary.

We want $X_t \xrightarrow{D} p$. The convergence theorem for finite irreducible Markov chains tells us that if Ω is finite, $pP = p$, and P is irreducible and aperiodic, then indeed $X_t \xrightarrow{D} p$.

Checking $pP = p$ is hard, as we have to sum over all Ω to evaluate pP. However...

Detailed Balance.
If there is a probability mass function $p_i, i \in \Omega$ satisfying $p_i \geq 0$, $\sum_{i \in \Omega} p_i = 1$ and

"Detailed Balance (DB)": $p_i P_{i,j} = p_j P_{j,i}$ for all $i, j \in \Omega$,

then $p = pP$ so p is stationary for P.

Detailed balance is sufficient for stationarity, and much easier to check. A Markov chain satisfying DB is reversible.

Convergence and the Ergodic Theorem

If $X_t \xrightarrow{D} p$, then $X_t \sim p^{(t)}$ with $p^{(t)} \simeq p$ at large t, so "most" of the samples $X_0, X_1, ..., X_T$, are "nearly" distributed like p.

For $f : \Omega \to R$ let $\hat{f}_T = T^{-1} \sum_{t=1}^{T} f(X_t)$.

Theorem. If $\{X_t\}_{t=0}^{\infty}$ is an irreducible and aperiodic Markov chain on a finite space of states Ω satisfying detailed balance with respect to the probability distribution p, then as $T \to \infty$

$$P(X_T = i) \to p_i \text{ and } \hat{f}_T \xrightarrow{a.s.} E_{X \sim p}(f(X))$$

for any bounded function $f : \Omega \to R$.

Such a chain is ergodic with target distribution $p_i, i \in \Omega$.

Metropolis-Hastings Algorithm

Target $p_i, i \in \Omega$ with Ω a finite set. The algorithm simulates X_{t+1} given X_t. This determines $P(X_{t+1} = j | X_t = i)$ and P.

If $X_t = i$ simulate X_{t+1} by accepting or rejecting $j \sim q(\cdot|i)$. This is called the "proposal distribution".

Let $X_t = i$. X_{t+1} is determined in the following way.

[1] Draw $j \sim q(\cdot|i)$ and $u \sim U[0,1]$.

[2] If $u \leq \alpha(j|i)$ where

$$\alpha(j|i) = \min\left\{1, \frac{p_j\, q(i|j)}{p_i\, q(j|i)}\right\}$$

then set $X_{t+1} = j$, otherwise set $X_{t+1} = i$.

We initialise this with some $X_0 = i_0, p_{i_0} > 0$ and iterate for $t = 1, 2, 3, ...T$ to simulate the samples we need.

Example: Simulating the hypergeometric distribution

$\mathrm{HyperGeom}(k; K, N, n)$ with $K = 10, N = 20, n = 10$ gives the probability for k successes in n draws without replacement from a population of size N containing K successes.

If $p_k = \mathrm{HyperGeom}(k; K, N, n)$ then for $k \in \Omega$,

$$\Omega = \{k \in \mathbb{Z} : \max\{0, n + K - N\} \leq k \leq \min\{n, K\}\}$$

Let $B^- = \max\{0, n + K - N\}$ and $B^+ = \min\{n, K\}$.*.

The PMF is

$$p_k = \binom{K}{k}\binom{N-K}{n-k}/\binom{N}{n}, \quad k \in \Omega$$

Give a MH MCMC algorithm ergodic for p.

*In our example $B^- = 0$ and $B^+ = 10$

Step 1: Choose a proposal distribution $q(j|i)$ that's easy to simulate and irreducible. A simple distribution that 'will do' is

$$q(j|i) = \begin{cases} 1/2 & \text{for } j = i \pm 1, \\ 0 & \text{otherwise.} \end{cases}$$

It's easy to simulate and irreducible: we can get from any state $i \in \Omega$ to any other state $j \in \Omega$ by adding or subtracting 1's.

Notice the proposal can lie outside the state space we defined above, $\Omega = \{B^-, B^- + 1, ..., B^+\}$. If $i = B^+$ and we propose $j = i + 1$ then j has zero probability in the target distribution.

One transparent way to deal with this is give these states probability zero in the target, setting $p_j = 0$ for all $j \notin \Omega$.

Step 2: write down the algorithm.

If $X_t = i$, then X_{t+1} is determined in the following way.

[1] Simulate $j \sim U\{i-1, i+1\}$ and $u \sim U[0,1]$.

[2] If $B^- \leq j \leq B^+$ and

$$\begin{aligned} u &\leq \min\left\{1, \frac{p_j\, q(i|j)}{p_i\, q(j|i)}\right\} \\ &= \min\left\{1, \frac{\binom{K}{j}\binom{N-K}{n-j}}{\binom{K}{i}\binom{N-K}{n-i}}\right\} \end{aligned}$$

then set $X_{t+1} = j$, else (if either condition fails) set $X_{t+1} = i$.

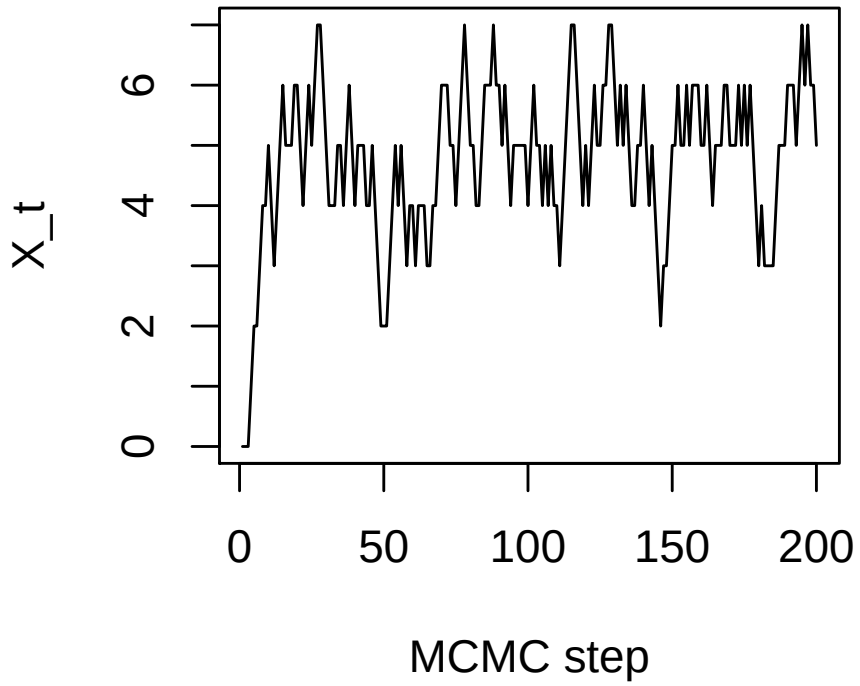
If $j < B^-$ or $j > B^+$ then $p_j = 0$ so $\alpha = 0$. Reject, stay in Ω.

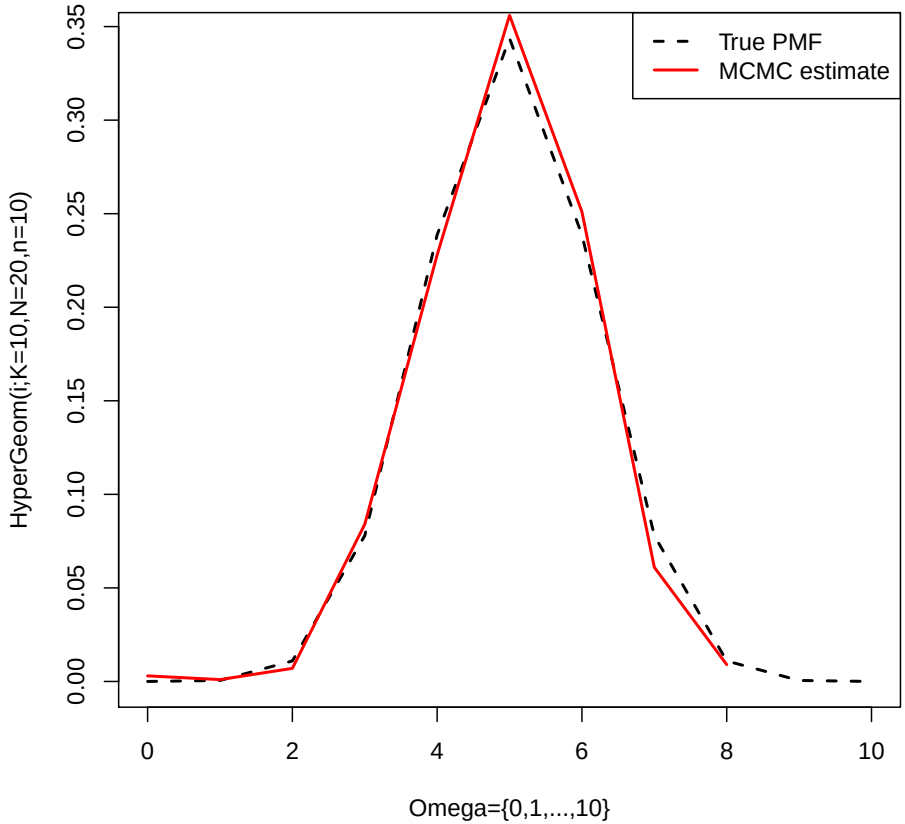
Step 3: check chain is irreducible and aperiodic. Clear here.

Left: x-axis is step counter $t = 1, 2, 3...200$. The y-axis is Markov chain state X_t targeting

$$\text{HyperGeom}(K = 10, N = 20, n = 10).$$

Right: histogram of $X_1, X_2, ..., X_T$ for $T = 1000$.





Ergodicity proof for Metropolis Hastings

Assume the chain is irreducible and aperiodic and show the chain determined by the MCMC update has stationary distribution p.

Compute the transition matrix P and verify detailed balance,

$$P_{i,j}\ p_i = P_{j,i}\ p_j,$$

since that implies stationarity, $p = pP$. DB for $i = j$ is trivial.

Suppose $j \neq i$. If $X_t = i$, then the probability $P_{i,j}$ to move to $X_{t+1} = j$ at the next step is

$$P_{i,j} = P(X_{t+1} = j | X_t = i) = q(j|i)\, \alpha(j|i).$$

Now check DB:

$$
\begin{aligned}
p_i P_{i,j} &= p_i \, q(j|i) \, \alpha(j|i) \\
&= p_i \, q(j|i) \min\left\{1, \frac{p_j \, q(i|j)}{p_i \, q(j|i)}\right\} \\
&= \min\left\{p_i q(j|i), p_j q(i|j)\right\} \\
&= p_j \, q(i|j) \min\left\{\frac{p_i \, q(j|i)}{p_j \, q(i|j)}, 1\right\} \\
&= p_j \, q(i|j) \, \alpha(i|j) \\
&= p_j P_{j,i}
\end{aligned}
$$

and we are done.

MCMC for continuous distributions

The candidate state $\theta' \sim q(\theta'|\theta)$ has distribution $q(d\theta'|\theta)$.
MCMC targeting $p(\theta),\ \theta \in \Omega$: Let $X_t = \theta$.

[1] Draw $\theta' \sim q(\cdot|\theta)$ and $u \sim U[0,1]$.
[2] If $u \leq \alpha(\theta'|\theta)$ where

$$\alpha(\theta'|\theta) = \min\left\{1, \frac{p(\theta')\, q(\theta|\theta')}{p(\theta)\, q(\theta'|\theta)}\right\}$$

then set $X_{t+1} = \theta'$, otherwise set $X_{t+1} = \theta$.

The rejection probability when the current state is $X_t = \theta$ is

$$c(\theta) = 1 - \int_\Omega \alpha(\theta'|\theta)\, q(d\theta'|\theta)$$

The transition distribution $K(\theta, A) = \mathbb{P}(X_{t+1} \in A | X_t = \theta)$ is

$$K(\theta, d\theta') = \alpha(\theta'|\theta)\, q(d\theta'|\theta) + c(\theta)\delta_\theta(d\theta').$$

The delta function $\delta_\theta(d\theta')$ satisfies $\int_A \delta_\theta(d\theta') = \mathbb{I}_{\theta \in A}$.

When $\theta' \neq \theta$ the transition distribution has a simple unnormalised density or kernel

$$K(\theta, \theta') = \alpha(\theta'|\theta)\, q(\theta'|\theta).$$

Rejection with $\theta' = \theta$ is automatically symmetric so DB is

$$K(\theta, \theta')\, p(\theta) = K(\theta', \theta)\, p(\theta').$$

Exercise: verify DB for K given above. Similar to discrete case.

MH example: an equal mixture of bivariate normals

$$p(\theta) = \frac{1}{4\pi} \sum_{i=1}^{2} |\Sigma_i|^{-1/2} e^{-(\theta-\mu_i)^T \Sigma_i^{-1} (\theta-\mu_i)/2}$$

with $\theta = (\theta_1, \theta_2)$. Use $\mu_1 = (1,1)^T$, $\mu_2 = (4,4)^T$ and $\Sigma_1 = \Sigma_2 = I_2$ for this illustration.

Step 1. For a proposal distribution q we want something simple to sample. A simple choice that "will do" is

$$\theta_i' \sim U(\theta_i - a, \theta_i + a), \ i = 1, 2$$

with $a > 0$ a fixed constant. We jump uniformly in a box of side $2a$, easy to sample, and

$$q(\theta'|\theta) = q(\theta|\theta') = \frac{1}{4a^2} \mathbb{I}_{|\theta_1 - \theta_1'| < a, |\theta_2 - \theta_2'| < a}.$$

Step 2. The algorithm is, given $\theta^{(t)} = (\theta_1, \theta_2)$,

[1] for $i = 1, 2$ simulate $\theta_i' \sim U(\theta_i - a, \theta_i + a)$

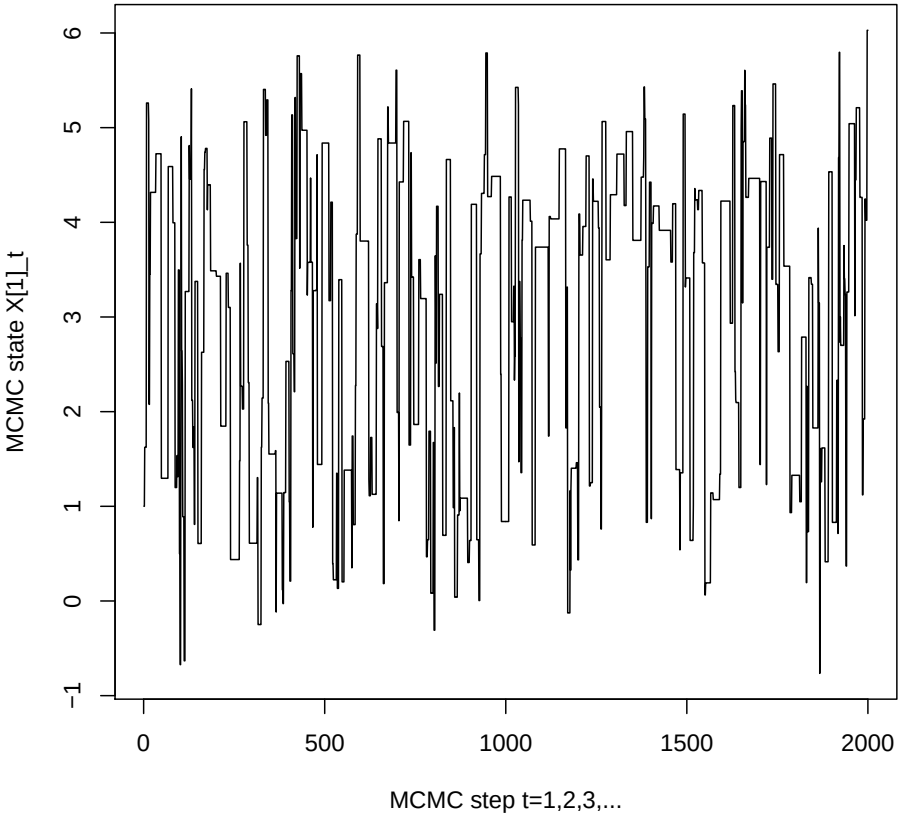
[2] with probability

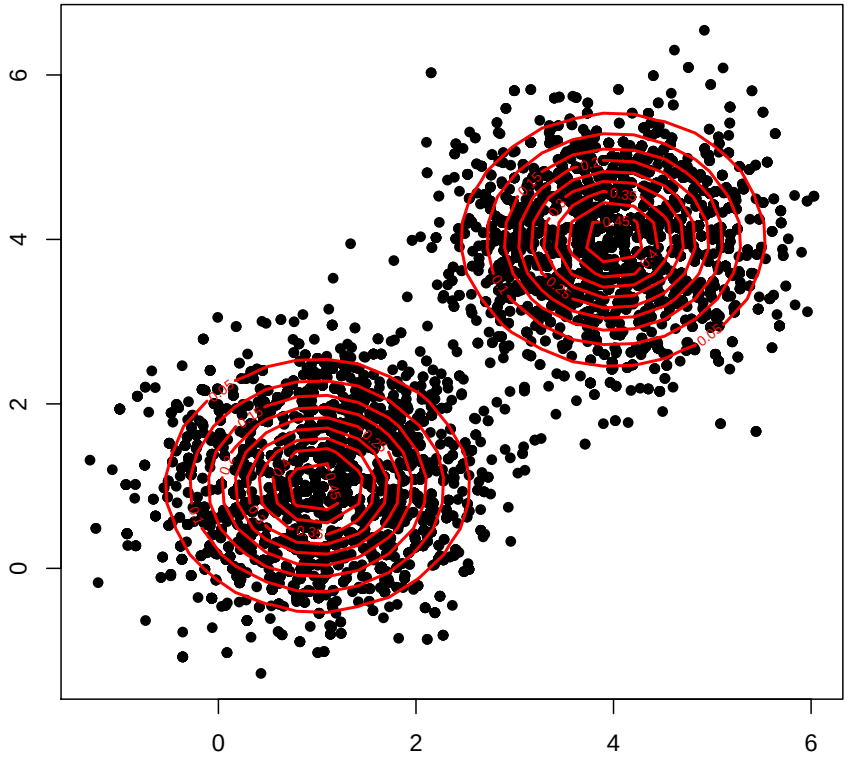
$$\alpha(\theta'|\theta) = \min\left\{1, \ \frac{p(\theta')}{p(\theta)}\right\}$$

set $\theta^{(t+1)} = \theta'$ otherwise set $\theta^{(t+1)} = \theta$.

Step 3. This algorithm is ergodic for any $a > 0$ but we will see that the choice of a makes a difference to efficiency.

Exercise: (see the associated R-file) try running this, experimenting with different values of a.


MCMC state X[1]_t
MCMC step t=1,2,3,...
−1
0
1
2
3
4
5
6
0
500
1000
1500
2000


0
2
4
6

Mixing updates for multivariate targets

If $\theta = (\theta_1, ..., \theta_p)$ and we target $p(\theta),\ \theta \in \mathbb{R}^p$, update one variable at a time. Small changes have higher acceptance probability.

Choose a component $i \sim \xi$ with probability $\xi_i,\ i = 1, \ldots, p$, set $\theta'_{-i} = \theta_{-i}$ and $\theta'_i \sim q_i(\cdot|\theta)$. The acceptance probability is

$$\alpha_i(\theta'|\theta) = \min\left\{1,\ \frac{p(\theta')\, q_i(\theta_i|\theta')}{p(\theta)\, q_i(\theta'_i|\theta)}\right\}$$

and the (unnormalised) transition density for $\theta' \neq \theta$ is

$$K(\theta, \theta') = \sum_i \xi_i K_i(\theta, \theta'),$$

where $K_i = q_i(\theta'_i|\theta)\, \alpha_i(\theta'|\theta)$. A mixture of updates satisfying DB also satisfies DB so the chain targets $p(\theta)$.

Example: normal mixture one variable at a time

Let $\theta^{(t)} = (\theta_1, \theta_2)$. Take $\xi = \left(\frac{1}{2}, \frac{1}{2}\right)$ and $q_i(\theta_i'|\theta) = \frac{1}{2a}\mathbb{I}_{|\theta_i' - \theta_i| < a}$.

1. Simulate $i \sim U\{1, 2\}$, $\theta_i' \sim U(\theta_i - a, \theta_i + a)$ and set $\theta'_{-i} = \theta_{-i}$.
2. With probability

$$\alpha_i(\theta'|\theta) = \min\left\{1, p(\theta')/p(\theta)\right\}$$

set $\theta^{(t+1)} = \theta'$, otherwise set $\theta^{(t+1)} = \theta$.

